Letter to Zalamea

The pitch

- The pitch

Introduction

- My relation to Z
- This letter: ready for the big book, appreciation for the epistemological significance of the ideas I humbly represent
- Why: Can finally read Lautman [1]
    - The perspectivist metaphysics of contexts as windows of opportunity for the infinite to grow via communicating communities encouraged to map their context [2]
    - The core of the solution: synthetic/analytic decomposition of knowledge
- My work as unraveling a thread from the core (an exercise) to a) the metaphysics/"theory of everywhere", b) borrow-work-replay/permacultural logic, c) the Hoapa implementation, d) the Hoapa conjecture (that b -> a is in c) [5]
- My path forward

0. Pre-historical and pre-philosophical intuition

- The movement of Lautman
- Pre-historical: how to think of the context we work with
- The methodology of Grothendieck
- Pre-philosophical: how to motivate our working hypothesis
- Hoapa as a generalization from math to science & pholosophy, from the individual to the collective

1. The map of the conjecture explored through authors and concepts

- The map of the conjecture [3, 4]
    - in category theory
    - in words
- Anchors [6]
    - Peirce
        - Community of Inquiry ≅ Hoapa
        - Agape ≅ G-evol
    - Lautman
        - The 'metaphysics → effective theories → reality' movement, where Hoapa is an effective theory solving the universe's creativity problem
        - The core solution as dual decomposition implemented as separation of concerns
        - Metamathematics of Hilbert as 'replay'
    - Grothendieck
        - The relative point of view as perspectivist-friendly mathematics, in synch with the universe (against the "view from nowhere" usual paradigm)

        - The rising methodology as respecting the separations of concerns (no hammer & chisel) and targeting integrations (patience and naturality of definitions)
    - Lawvere (Hegel, Grassman, Kock)
        - Cohesion,
        - Synthetic geometry as the free/thorough variation of the conceptual plane, separated from the fixed words of the UTL graph
    - Deleuze
    - Simondon & Kauffman
        - Individuation process as the implementation of permacultural logic: do not exhaust your milieu
        - Autocatalytic sets as biological implementation
- Grounds
    - Borrow-work-replay
    - Intensive/extensive
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2. Separations of concerns and integrations translated in model/set theoretic logic

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Project Report

The pitch

Start with how one can combine notions in the philosophy of mathematics with ideas from the engineering of networks, to think an extension of the scientific method to the collective which incorporates participative democracy, or more generally a theory of work. And recognize both what it can do, and its place in Nature.

The context

I started my journey into epistemology as the concerned citizen I've always been, but by chance, when I stumbled upon an interesting question, or rather an exercise: what is the most universal and balanced communication system possible? (And how to make this question rigorous.)

I call this system, and the whole project of understanding it in depth, Hoapa.

In this open letter addressed to Fernando Zalamea, I will present the point of view on the construction emanating from those specifications which best exhibits how important it could be, both as a *product* of science, and to *itself*, the scientific activity.

While my communication around my work has focused on the direct action prescribed by the latter, in this text I will embrace my role in the development of the former. Essentially, I'm now ready for the "big book" and 10 years of solitary work to which Zalamea urged me to commit.

The turning point for me has been the comprehension of Albert Lautman, after many perplexing attempts. In particular the movement he sees between dialectical or metaphysical problems (which can justify the

construction), and solutions to those found in effective mathematical or physical theories (which resemble the construction).

The first part of the letter thus considers Hoapa through a series of dualities: concrete & abstract, synthetic & analytic, individuality & variation, in a context regulated by a tension between structure and genesis.

In the second part we broaden the epistemological scope, and revisit those concepts through the different authors who also gave them life in their work: Alexander Grothendieck, Bill Lawvere, Gilbert Simondon, Stuart Kauffman, Gilles Deleuze, Charles S. Peirce. Using those I will reconstruct the state the project is currently in: a conjectural adjunction between Hoapa as a process and some category representing "nice mathematics".

Third I will translate the design of the construction in the language of the Helsinki Logic Group, where I will present my work in the fall of 2026. This adaptation into model and set theoretic logic should demonstrate that our seemingly esoteric considerations do produce interesting and useful structures for the working mathematician.

I will also try and convince this audience that the logic they know and love is implicitly *ecologist*, that its structural laws participate in a universe designed for sustainability and solidarity.

The name, Hoapa, comes from one of the many San languages and means "everywhere", as an homage to this culture, now largely extinct, who are the proud descendants of the first implementers of absolutely universal and balanced communication systems, masters of sustainability and solidarity.

Hoapa is too humble and respectful of difference to be a theory of everything, but I believe its network shape and resilience allows it to be a theory of everywhere, true in every context.

0. Hoapa as an implementable architecture

1. Hoapa from Lautman

Hoapa is designed to exist in a specific context, which we can think of naively or pre-philosophically as a world in which there are concepts, some of these concepts being real, some of these things being alive, some of those organisms being capable of complex thoughts and some of them able to communicate those thoughts in details. If we consider that being alive is like playing a game, then there is an invariant good strategy for organisms which consists in mapping one's context, in gathering as much information as possible about one's circumstances and finding a representation and processing scheme for it. In the evolution we know this gives brains (and later computers). In the subgroup capable of detailed language the main strategy remains the same, but the possibilities awarded by communication broaden the range of efficient implementations, and shift the scale of operations one level up. Hoapa is one of those specific implementations.

Lautman allows us to determine Hoapa from a more general context, the one he worked with: metaphysics, and to follow his movement from dialectics to effective theories. Essence & existence, whole & part, continuous & discrete… are what he calls *notions*, usually organized in pairs, whose relations and tensions are *Ideas*. They *dominate* mathematics and physics, since the theories of the latter "only" exist as effective solutions to the tensions of the former (prior versus secondary, but without involving temporality). Lautman sees the signs of this movement in the recurrence of structure and content in mathematical theories which would otherwise be unrelated, were it not for their common dialectical origin (ex: local/global in sheaf

cohomology, Galois descent, gauge field theory). This will later be made rigorous, at the effective end, through category theory.

1.1 Concrete & abstract

The tension presiding over Hoapa, seen as an effective theory resolving a dialectical problem, is between the concrete and the abstract. Here the relation is straightforward: mutual exclusion and copy. It proceeds from the simple observation that a map can't include the things it's supposed to represent, or that the narrator of a story shouldn't interfere with it. The world to be reflected occupies all of the concrete, the reflecting apparatus only a small fragment of it, and it's urged to get its work done in the abstract realm. And yet, from the reflector's point of view, it's the reflected world which becomes the abstract, and it is the concrete.

In this sense Hoapa is a close sibling to the theory of covering spaces, because of the obvious copycat relation, but also in how they must preserve this reflection even in their own transformations. For instance: a covering $p : E \to B$ provides a space in which distinctions between lifts become available, while its deck transformations g preserve reference to the base: $p \circ g = p$. That illustrates transformations which preserve the specified projection. Hoapa's analogous ambition is to enable traceable work while keeping untouched the relation between participants and their context, between abstract work and the concrete context it works on. This *separation of concerns* will give us the next dialectical tension, internal to the reflecting side, between two kinds of decompositions for knowledge: essentially, brains and texts.

At this "metaphysical surface level", we can thus see the interior of Hoapa unfolding, or we could say we hear it protesting and objecting to the premature closure. But if we look at how Lautman structured his main thesis, we can also peak at what will become in our story the open enveloping network, organized according to further dialectical tensions, in which the thing we talk about is one instance among countless many. In his example of Riemann surfaces, passing to the universal covering surface enables new constructions of functions. He describes the resulting connection between structure and genesis as follows:

> [T]he ascent towards the absolute has as a consequence, not only to confer on a mathematical entity the greatest simplicity of internal structure possible, but to make it able to give rise to entities other than itself.
> Albert Lautman, Mathematics, Ideas and the Physical Real (Continuum, 2011), pp. 130–131.

This passage is at the hinge between parts 1 and 2 of his primary thesis, "The Schemas of Structure" and "The Schema of Genesis". This dialectical tension is where Hoapa can ultimately participate in, but only after some work towards its own closure.

(I remark in passing that Hoapa's surface and first structuring tension has also been chosen by Zalamea to structure his introduction to Lautman's texts, with similar emphasis on coverings, sheaf and descent, the Galois correspondence.)

{which starts here with a *separation of concern*}.

This term, coming from software engineering, is synonymous to modularity.

(like your brain changes at a much higher rate than your worldview
His main point, or structural idea,
That illustrates transformations which preserve the specified projection (Hatcher, §1.3).

The Lautmanian category theory preserves this invariant in effective theories: logic, coverings .

Levels.
Zalamea -> concrete / abstract
Core -> synth / anal (start synthetic brains by recalling that they must be insulated?)
Concepts -> variations / individuation

- Lautman -> structure / genesis

a covering space projects onto a fixed base, reproducing its topology locally while potentially unfolding global identifications. The universal cover, when it exists, carries this unfolding to a simply connected space.

keeping a trace of how participants are in

articulation and transformation while preserving declared relations to context, and to propose a corresponding requirement for the conditions of participation: participants' standing, and the boundary between their lives and their contributions.

{What's interesting: Hoapa creates its own concreteness

2. Hoapa in Grothendieck, Lawvere, etc.

3. Hoapa in model and set theoretical logic

Conclusion

Codex

Appendix — Letter to Fernando Zalamea: how Lautman now organizes Hoapa

23 October 2026. Dried-out content inventory for a letter, not the letter itself. Focus: the intellectual change, the biological and semiotic passage into it, the resulting series of architectural demands, and the question addressed to Zalamea.

1. The news to give him

- **Opening:** Lautman is no longer only a philosophical justification attached to Hoapa. His problematic relations now help organize its construction and expose what remains to be built.
- **Change of emphasis:** from proposing a communication protocol to investigating the organization of collective understanding as a form of work.
- **Distinctive extension:** mathematical inquiry produces effective theories; Hoapa proposes to make the collective organization of that inquiry itself the subject of an effective mathematical theory.
- **Personal transition:** return to his encouragement to undertake the long work. Explain what has finally made that work intelligible: a construction whose parts can be read through connected problems, rather than a promising protocol surrounded by a growing bibliography.
- **Core question:** how can a community differentiate its understanding of a world, give its achievements a determinate and reusable form, and enable further inquiry without turning either people or

knowledge into fixed instruments?

{Lead with what changed in your work, not with an exposition of Lautman to his interpreter. “Natural in Lautman” should become visible through the construction. It need not be announced as a verdict he is asked to accept.}

2. What to take as shared ground, what to introduce

- **Shared ground:** Lautman's dialectical problems and effective realizations; Zalamea's structural and dynamic reading; mathematical transit, mixtes, and the relation between structure and genesis. Recall the particular passages doing work, not their author's whole position.
- **Peirce belongs on the shared side.** Zalamea's own work on continuity and existential graphs makes him an interlocutor for the semiotic connection, not a reader who needs Peirce introduced. The news is how Hoapa now uses that connection. Preparation reference: Zalamea, Peirce's Logic of Continuity.
- **Introduce briefly:** the specific Kauffman work–constraint cycle; Montévil–Mossio's constraint closure; Pattee's epistemic cut and semantic closure; the exact Simondonian problem being connected to them.
- **Explain fully enough:** Hoapa's common participant interface, differentiated documentation, variation responsibility, and passage from completed work to further work. Familiarity with the philosophers does not supply familiarity with these design choices.
- **Keep in reserve:** runtime details, Agda module names, alternative nerve architectures, and the entire anchor catalogue. One scoped mathematical example is more useful here than a status report.

{This is an audience strategy, not a claim about what he has or has not read. His familiarity with Lautman and Peirce is documented; his familiarity with these particular biological arguments should not be presumed either way.}

3. The entry through work: what the letter must supply

- **Kauffman's question:** not only what an activity produces, but what channels it and what produces or maintains those enabling constraints.
- **Atkins, through Kauffman:** energy release becomes directed work through constraints. A cannon's bore supplies the familiar example; constructing the bore requires other work. Kauffman, p. 8.
- **Biological turn:** the organization participates in maintaining the conditions of its own activity. Montévil–Mossio distinguish transforming processes from constraints stable on relevant timescales; closure concerns their interdependent maintenance. Material and energetic openness remain. Closure of constraints.
- **Question carried into Hoapa:** what lets an achievement become an enabling condition for subsequent inquiry? Who or what keeps its interpretation, access, and further use possible?
- **Why this is more than an organism metaphor:** the object of comparison is an organization of enabling relations. Human participants do not become organs defined by their service to it.

{For this letter, the work–constraint cycle is sufficient. Keep autocatalytic, work-task, and constraint closure distinct, but do not reproduce their whole taxonomy. None currently proves that Hoapa is alive, self-maintaining, or sustainable.}

4. Simondon, Pattee, Peirce: the bridge, not a lineage

Simondon supplies the process question.

- Begin with individuation, not an already complete individual placed in an independently complete environment. Individual and milieu emerge through the resolution of a problematic situation; further potentials remain. Primary orientation: Simondon's summary of his thesis, reproduced on the family-maintained site.
- Proposed biological reading: Kauffman and the closure-of-constraints account make some of these questions operational—how an organization arises, supports its activity, and depends on the milieu from which it differentiates.
- **Your "biological implementation of Simondon":** a candidate biological realization of part of that problematic. Closure makes continued activity intelligible; Simondon also asks how individuation transforms the organization and its problems.
- Hoapa needs both: maintenance of an inquiry's conditions; genesis of achievements that can reorganize those conditions. Maintaining a cycle is not yet explaining renewal.

Pattee supplies the symbolic detour.

- Distinguish dynamical processes from symbolic descriptions and constraints. Both are physically realized; their descriptions perform different jobs.
- A symbol operates through interpreting machinery. Semantic closure concerns the reciprocal dependence of descriptions and the organization that realizes them. The Physics of Symbols, especially §10.
- Proposed extension for the letter: activity → description → reasoning or design → interpreted activity; activity also maintains the means of describing and interpreting.
- **"Pattee-loop" names our use of this distinction.** It makes the concrete/abstract passage inspectable; it is not Pattee's name for a universal algorithm of thought.
- The abstract detour changes what work can be organized: retain a description, compare alternatives, communicate a construction, use it again. It does not make symbols immaterial or their manipulation universally cheap.

Peirce supplies the epistemological return.

- Reconnect this detour to sign, object, interpretation; clarification through conceivable consequences; inquiry capable of changing habits and subsequent conduct. How to Make Our Ideas Clear.
- What the biological and symbolic accounts add to Hoapa's use of Peirce: explicit questions about the material organization that carries interpretation and sustains its continuation.
- What Peirce adds in return: a description matters through interpretation and further inquiry, not merely because a mechanism stores or transmits it. A successful control loop is not automatically an account of meaning or truth.
- Collective translation: a result must be usable, examinable, and revisable by others. Documentation is an achievement in a continuing inquiry, not a terminal verdict owned by its author.

Division of work: Simondon — individuation with a milieu; Kauffman and Montévil–Mossio — biological enabling and maintenance; Pattee — symbolic mediation within physical organization; Peirce — meaning and inquiry; Lautman — the problematic relations read through their effective constructions.

{"Implementation" and "extension" name the proposed articulation in Hoapa. They do not establish that Kauffman implemented Simondon's philosophy, that Pattee derived his theory from Peirce, or that the

theories are equivalent. Leave the electrical-logic priority story out: it is unnecessary to this connection.}

5. Why the Lautmanian series begins with concrete / abstract

- **The initiating problem:** an activity inside a world devotes part of that world to describing and understanding it. The description must differ from its subject while remaining answerable to it.
- **The Pattee detour exposes the passage:** the apparatus, the description it carries, and the interpretation through which that description affects further activity. Distinguish these without separating them into causally isolated substances.
- **Hoapa's surface:** a community works on descriptions of a context; producing those descriptions does not entitle its protocol to govern the context or its participants.
- **Relative reversal:** descriptions and their production are themselves concrete achievements for another inquiry. For the inquiring community, the world is encountered through abstractions. Concrete/abstract are relative positions, not matter versus unreality.
- **Political consequence chosen by Hoapa:** participation, refusal, assent, and disagreement must not become levers of reward, penalty, or rank. Material support remains necessary; it must not purchase a verdict or authority over its producers.
- **The next problem appears inside the descriptive activity:** how should understanding differentiate without imposing corresponding kinds and ranks on the people contributing to it?

{This is the starting point of Hoapa's Lautmanian reading, not a claim that Lautman placed concrete/abstract first in a universal deduction. The political boundary is our architectural commitment; it is not a law inferred from the behavior of neurons or genes.}

6. The remaining tensions: how one opens the next

Synthetic / analytic — common participation, differentiated understanding.

- Subject decomposed: **understanding of the world**, embodied among participants and articulated in documentation.
- Lautman's dimensional/common-plan decomposition: the reference for a shared mode of contribution. Hoapa's participants have equal formal standing without being identical, interchangeable, or exhaustively represented.
- Lautman's proper/constituent decomposition: the reference for differentiated concepts, claims, dependencies, and evidence. The documentation specifies what the inquiry has articulated.
- Both moments occur in both realms; the architectural emphasis differs. A common actor interface is not a theorem about equal minds; a conceptual graph is not a canonical prime factorization.
- Two-way boundary: social rank must not settle a claim; a claim's evaluation must not settle its producer's rank. Knowledge may specialize without compulsory specialization of people.
- **Transition:** differentiated documentation needs comparisons and stable names, but those names must not erase the possibilities of what they name.

Variation / individuation — determinate achievements, open possibilities.

- A concept must become determinate enough to teach and reuse. Its stabilized presentation need not exhaust its families, responses, alternative realizations, or extension problems.
- Mathematical resources: witnessed identifications, groupoids and moduli, deformation theory, Kock–Lawvere variation. They offer different exact treatments, not one already constructed Hoapa layer.

- Small example: two descriptions agree at one parameter value and disagree in first-order response. They may be equivalent for reporting that value but not for continuing the investigation.
- Universal-language ambition: reusable conceptual articulations whose comparisons, operations, and relevant variation travel together. The “word” is not the obliteration of everything surrounding its stable presentation.
- **Transition:** a differentiated achievement can supply a new common plan. Documentation becomes synthetic for later inquiry; a product becomes part of a context.

Structure / existence and genesis — the overarching question.

- Ask what constructions the organization makes possible, and what its completed constructions can enable in turn. This traverses the preceding pairs rather than adding one more compartment.
- Lautman's ascent toward structural simplification and new constructions supplies the mathematical orientation. Simondon keeps the achieved individual related to its milieu and continuing individuation; Peirce keeps the result related to further interpretation.
- **Borrow–work–replay:** use situated conditions; perform inquiry; make its achievement accessible; let others take it up under stated conditions. Completion of an episode need not complete its subject.
- Distinguish a reusable result from a reusable rule; maintenance from novelty; a new object from changed contextual capabilities. Neither repeated activity nor a new label establishes genesis.
- **Permacultural question:** can this work preserve achievements while sustaining or renewing the conditions of further inquiry, without consuming the freedom of its participants?

The series to retain: world/description → common participation/differentiated understanding → determinate concepts/open variation → achievements becoming conditions of further work. Structure/genesis asks what the whole organization enables.

{The arrows record the path of problems and design obligations, not proved entailments. This is where the letter should be most concrete: show what each tension demands of the protocol. Do not substitute the list of pairs for their architectural consequences.}

7. What makes this a mathematical programme, not only a reading

- **Separation and integration:** distinctions permit legitimate compositions; integrating settings reveal further distinctions. The relation is reciprocal, not one side fragmenting and the other erasing differences.
- **First standard instance:** descent—restrict, compare on overlaps, assemble when compatible. An incompatibility can instead become a faithfully documented result.
- **Second bounded problem:** a representation can preserve today's answer yet lose tomorrow's operation. Ask whether the operation factors through the retained description; §2.11 gives the finite example and its proof.
- **What these examples buy:** something the proposed interpretation must preserve, and a case in which it fails. They do not prove the full architecture optimal.
- **Conjectural horizon:** a relation between organized practices K and contextual structures M, provisionally Allow ⊣ Require. K's separations and M's integrating resources should constrain each other's mathematical specification.
- **Zalamea connection:** his passage between determinate constructions and freer structural relations motivates this correspondence. Both sides would still be effective mathematical structures; M is not identified with Lautman's realm of Ideas.

- **Specialness sought:** adequacy of an organization to a context and to further contextual development, under independently stated requirements. Not “Hoapa is best” independently of what it is for.
- **Status in one sentence:** finite locality and resource-sensitive structures are partly checked; the cross-side adjunction and the broader adequacy theorem remain to be constructed.

{Use a short recall of his structural/dynamic Platonism, not the long quotations already collected in §5.1. The distinction between conjecture and construction should occupy one clear sentence, not take over the letter.}

8. The question and invitation to end with

- **Main request:** does this extension preserve the force of Lautman's movement from problems to effective theories when the subject is the collective activity producing such theories?
- **Specific interpretive test:** are the two decompositions doing legitimate work when their common subject is understanding, distributed between embodied participation and differentiated documentation?
- **Specific bridge to discuss:** does the Simondon–biological organization–symbolic mediation–Peirce articulation clarify the concrete/abstract problem, or does one passage require a different notion?
- **Mathematical direction:** which small construction would best expose the strengths or limits of this reading? Offer the finite continuation problem as one candidate, not a demand that he audit the formalization.
- **Personal continuation:** this is the point from which the long work he encouraged now seems possible. Ask for a critical orientation, a source, or an interlocutor—not certification of the complete project.

Letter order: the intellectual news; the brief work/symbolic detour; concrete/abstract and its reversal; the two internal tensions; genesis and the conjectural horizon; one precise request. Keep the full report as supporting material.

Leave out unless requested: the full history of M's revisions; the implementation ledger; all three closure definitions; a general theory of life; the complete anchor constellation; claims of priority for Peirce; a categorical proof of political commitments. Preserve the ambition by making its path legible, not by compressing every branch into the letter.

Preparation locators: the report's §§3–5 above; arc/22, L6079, “Digest — three pairs”, and L6092, “Concrete | abstract”; arc/23, L7, “definition of work”, L27, “thinking”, and L159, “Could we say that Peirce”. These are records of the developing interpretation; the primary-source links above support the narrower author attributions.

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Claude

Appendix B: letter to Zalamea — elements, dried

Task: convey how Hoapa **now** uses Lautman, and that the series of dialectical tensions **begins** with concrete/abstract, entered through Kauffman on work, extended into a Pattee-loop, which is the biological implementation of Simondon and the extension of Peirce.

Calibrated to him throughout: explain nothing he owns, introduce in one line what he does not.

Calibration: what he already has

Do not explain, do not summarize, use as shared vocabulary:

- **Lautman entire.** Notions and Ideas-as-problems; *structure et existence*; schemas of structure and of genesis; mixtes; solidarity; the double movement; the priority of the problem in the order of reasons.
- **His own apparatus.** Transit; saturation and liberation; the *triple va-et-vient*; residues and obstructions; eidal / quiddital / archeal.
- **Peirce in depth.** *Peirce's Logic of Continuity*. Pragmaticism, synechism, triadic semiosis, existential graphs, agape, the community of inquiry.
- **Grothendieck.** Relative point of view, rising sea, descent, the *Esquisse*, dessins, Galois–Teichmüller.
- **The mathematics.** Sheaves, descent, Galois correspondences, coverings, the structural ladder of his own long passage.
- **Deleuze on Lautman.** Including the transformation of dialectical Ideas into immanent problematics.
- Riemann, Galois, Grothendieck as his three pillars.

Calibration: what he does not have, one line each

- **Atkins' definition of work** — the constrained release of energy into a few degrees of freedom. Not force through a distance. Energy counts as work only when a constraint channels it.
- **Kauffman** — the work–constraint cycle: "work begets constraints beget work" (*Investigations*, 2000). The cannon is the constraint, and it took work to make the cannon. Also autocatalytic sets, and non-ergodicity above the atomic scale, from which evolution is *historical*.
- **Montévil–Mossio (2015)** — constraint closure: A constrains the release in process 1 that builds B, B constrains 2 that builds C, C constrains 3 that builds A. The system does the thermodynamic work of building its own boundary conditions. Function is thereby naturalized: a part has a function iff it is a constraint maintaining the closure.
- **Pattee** — the *epistemic cut*: lawful, rate-dependent dynamical constraints against rate-independent **symbolic** constraints whose physical form is arbitrary relative to what they stand for. *Semantic closure*: the symbols specify the machinery that reads the symbols.
- **Energy-degeneracy** — the thermodynamic condition for a substrate to hold abstractions: its states must cost about the same whichever meaning they carry, so physics does not decide the content. Not Peirce's; a twentieth-century reading, Szilard and Landauer to Pattee.
- **Simondon** — probably a name rather than a working anchor. Preindividual field; metastability as a genuine third state; individuation as the progressive resolution of a local incompatibility, structuring individual **and** milieu together; transduction; the reserve is not exhausted by a successful individuation.
- **Concurrency** — actor model: send, receive-and-update, create. Session types and duality $S \leftrightarrow S^{\perp}$. Not his field; keep to two sentences.
- **The formal layer** — cubical Agda; what is banked; `session-01-action` as the one named postulate.

The one door he already owns: biosemiotics

- Hoffmeyer, Emmeche and Deacon are **explicitly Peircean**; *Peirce and Biosemiotics* (Romanini & Fernández, Springer 2014) exists.

- So the whole Pattee / closure / code-duality complex can be entered through the Peircean door rather than introduced cold.
- **Most efficient calibration move in the letter.** Enter by his author, not by the biologists.
- Code duality (Hoffmeyer–Emmeche): an organism exists twice, as a digital message to itself across generations and as an analog body interpreting its world. Hoffmeyer's increasing *semiotic freedom*.

The spine, in one chain

1. **Peirce, the original figure.** Substrate-independent logic — the Marquand letter, 30 Dec 1886, logic by electrical switching. "Logical Machines" (1887): every machine reasons, since the relations among its parts entail relations nobody intended, but every machine has two impotencies — no originality, so it cannot find its own problems, and it can do only what it was contrived for. **Machines apply; they do not design.** "Pragmatism" (1907): brute-force action is dyadic, semiosis irreducibly triadic — Pattee's cut sixty years early. Same manuscript: the **energetic** interpretant is an exertion, which is work; the **ultimate logical** interpretant is a *habit-change*, a standing constraint on future action. So semiosis passes through work and ends by building a constraint. 1891–92: laws are habits the universe has acquired.
    - Qualify at once, because it is a claim about **his** author: Peirce is the original figure but not the root of the lineage. Jevons precedes him on logic machines (1869) and his priority is the *electrical* idea; the Marquand letter was published only in 1993; Shannon reinvented switching logic independently in 1937; von Neumann, Pattee, Kauffman and Montévil–Mossio do not build on him. The genuinely Peircean line is biosemiotics and Deacon. And the interpretant triad appears in one manuscript which Peirce later abandoned (Jappy 2024).
2. **Simondon, the philosophical account.** How an individual precipitates from a metastable field, structuring its milieu with it, leaving the field capable of further individuation. Transduction propagates it stepwise.
3. **Kauffman and Montévil–Mossio, the biological implementation.** Work as constrained release; the cycle; closure as the point where every constraint the system needs is produced inside it.
    - **The sentence for him:** a constraint is **structure**, work is the act that brings into **existence**, and the cycle is a genesis that loops — structure engenders existence as the constraint channels energy into work, existence engenders structure as that work builds the next constraint. So the work–constraint couple maps onto *structure et existence*, and the loop is a **schema of genesis**.
    - Two further pairs of his fit closure itself: **intrinsic against induced** — the cannon's constraints are induced by its maker, the cell's are intrinsic, and closure is where all the relevant ones become intrinsic; and **the ascent toward completion** — algebraic closure means every equation over a field finds its roots inside the field, constraint closure means every constraint the system needs is produced inside the system.
4. **The Pattee-loop, the extension into the symbolic.** Constraints become energy-degenerate, so their *structure* is freed from the cost of their *existence*. That freedom opens a gap between heterogeneous domains — and a **mixte** is exactly what occupies it. The ribosome and tRNA are mixtes between code and chemistry; perception and motor action are the two mixtes performing the flips.
    - **Lautman's own mathematical mixte closes the circle:** Hilbert space, a synthesis of continuous and discrete, where a function becomes a list of coefficients. That is the flip in mathematical form.
    - Why the detour is worth making, in Atkins' terms: **leverage** — a cheap constraint gates a large release; **memory** — a constraint stored abstractly is reusable without being rebuilt. The cannon

has neither, which is why it never makes the detour.
    - Thinking is therefore **cheap work on constraints that channel expensive work**.
    - For it to be closure rather than a controller sitting on top, the metabolic side must also do the work that rebuilds and maintains the abstract side.
5. **Concrete/abstract arrives as a problematic, not a stipulation.** An arbitrary piece of the world is dedicated to manipulating abstractions, which design and apply back into the concrete it started from. And what counts as concrete or abstract is **relative** — his own relativity, where a mathematical object is concrete relative to a more abstract one.
6. **Then the series.** concrete/abstract → synthetic/analytic → individuation/variation → structure/existence and genesis.

How Hoapa now uses Lautman: what changed

- **The entry point moved.** Previously Lautman entered through the two decompositions mapped onto the architecture. Now **concrete/abstract is primary**, entered biologically through work and closure, and the two decompositions are the *second* tension rather than the first.
- **The domination thesis is now carried by the recurrence argument**, in its correct form: the same *problem* recurs across theories with no common mathematical content — local/global in sheaf cohomology, in Galois descent, in gauge field theory — so what they share cannot be internal to any of them. Not recurrence of content, which would make the theories related and the argument circular.
- **The decomposition stays rigid inside K.** The Lautmanian problem lives **at the adjunction**, not inside either side. Rigid endpoints are the precondition for a liaison between them to be statable at all. Documentation's genuine aspiration to become synthetic — furnishing languages for later understanding — is then a **datum of the round trip** `Allow ∘ Require`, not a drift in the assignment.
- **Mixtes are used as a source question for adapters**, not as a classification: what must an intermediate structure retain from each setting to make the passage possible? Explicitly **not** an identification of the project's R4 adapter with a mixte.
- **Structure and genesis now supply the renewal criterion**, with the hinge passage between the two parts of the thesis doing the work.
- **A third tension has been added that is not in Lautman**: individuation/variation. Say this to him before he says it: the four-tension series is a *proposed Lautmanian interpretation of modern mathematics*, not a claim that Lautman formulated the hierarchy.
- **The reading is immanent, not orthodox.** Lautman gives the Dialectic an ontological priority over the mathematics realizing it; the project uses the Deleuzian and Corfieldian transformation into problematics immanent to practice, because a project giving collective inquiry a protocol cannot coherently hold that the problems enjoy priority over any practice. Say so, or he will assume the orthodox reading.

The four tensions, in order, with what is new in each

- **I. Concrete / abstract.** How particular realities and general structures determine one another. Entered through §3's work and loop. Lautman's pair here is *les systèmes formels et leur réalisation*, and his own example is the **Skolem paradox** — a formal system cannot pin down its realizations. Pattee's cut is that shape with the realization made physical.
- **II. Synthetic / analytic.** His two decompositions: **proper, individual, analytic** — prime factorization, the roots of a polynomial, from the particular toward constituents characterizing it; and **dimensional, synthetic** — a vector relative to a dimension and a chosen basis, a common plan supplied by the

ambient domain, expressing the **solidarity** of the elements. Not absolute: a polynomial factors over one field and not another, so extension of the base discloses constituents previously unavailable. Project reading: embodied understanding predominantly synthetic through one unityped mode of participation, documented understanding predominantly analytic. Both moments in both realms; neither M nor K is "the synthetic side". `propre` is also the French for *eigen*.

- **III. Individuation / variation.** Weighed against discrete/continuous (too general, misses decomposition) and rigid/deformable (does not express coarse-graining). Chosen, with rigidity and deformability as its principal mathematical manifestation. The ladder: eigenvalue decomposition, moduli spaces, persistent homology. A decomposition can stay exact while varying continuously; individuation of constituents can fail at singularities and degenerations; a coarse-grained invariant can stay stable throughout. **The third level reactivates the second**: families of decompositions become new synthetic wholes, which become objects of abstract investigation again.
- **IV. Structure / existence and genesis.** The hinge: *"the ascent towards the absolute has as a consequence, not only to confer on a mathematical entity the greatest simplicity of internal structure possible, but to make it able to give rise to entities other than itself."* His Riemann-surface example: passing to the universal covering surface enables new constructions of functions. Project use: the renewal criterion — specified earlier achievements remain recoverable while a genuinely unavailable continuation becomes admissible; and the sharp negative, that **different sites can present equivalent topoi, so a changed site alone establishes nothing**.

Meeting him on his own ground

- **Lead with coverings.** He structured his own introduction to Lautman around coverings, sheaves, descent and the Galois correspondence. §4.1's passage is a covering-space passage. His emphasis returning as the project's entry is the strongest single move available.
  - The passage, with its hedge intact: a covering `p : E → B` provides a space in which distinctions between lifts become available, while deck transformations preserve reference to the base, `p ∘ g = p` — transformation without change of the *specified projection*, not without change of anything whatsoever. Structural analogy, not mathematical identification; it neither derives the requirement for participation nor establishes that Hoapa satisfies it.
  - And the two productive differences: deck transformations form a **group**, while the invariance actually proved on the participant channel is over injective, not necessarily surjective, relabelings — a **monoid**; and a covering preserves one canonical projection while Hoapa would preserve a **declared family** of relations, which puts the nearer home at a site with a declared topology.
- **Use his annotation of his own two passages**, since it is the project's conjecture in his words: ① the world rich in determinations, full of precisions and delimitations = **K**; ② the world of ideas of possible, free relations = **M**; ③ the inevitable back-and-forth = the category the adjunction produces. Note the direction is the reverse of the naive guess — K is the determinate side.
- **His sentence doing the project's work:** restrictions and delimitations are an *enrichment* of knowledge through the precision and certainty they provide, not an impoverishment. Pair it with separations of concerns: each restriction buys a guarantee.
- **Solidarity**, his emphasis and Lautman's word. Four senses: between whole and parts; between a mathematical being and its ambient universe; between domains and operations; reciprocal between theories. Solidarity is the relation, genesis the movement.

- **Saturation and liberation** attributed to *his* reading of Lautman. Then the project's non-conflation: K's technical *causal saturation* — every globally witnessed causal relation between a conversation's events is already generated internally to it — is a different predicate with an unfortunate name overlap, and is not Simondonian metastability either.
- **The licence for the biological extension is Deleuze**, whom he knows on this better than anyone: *Difference and Repetition* takes Lautman's Ideas-as-problems and applies them to the organism, with genes and development as the actualization of an Idea. Lautman himself worked almost entirely within mathematics, with occasional forays into physics. Cite the precedent rather than assert the extension.

What to ask him

Precise questions, each answerable only by him.

1. **Has anyone related Lautman to Peirce?** He is the major living scholar of both. If the answer is no, that is the project's opening; if yes, it is a literature the project needs. Probably the single best question in the letter.
2. **Does the four-tension series map onto the *triple va-et-vient*, and in what order?** Connecting them rigorously needs the actual wording and sequence of his three movements, which the project has not done.
3. **Where exactly does Lautman use "solidarity", and does it bear this weight?** Four senses have been located; whether they are one notion is his question.
4. **Is the biological extension licit beyond Deleuze's precedent?** Did Lautman ever go past physics?
5. **Structure/existence against essence/existence** — which does he take as the primary pair, and does the distinction bear on the schemas of genesis?
6. Optional, and the most useful if he bites: **is the passage from an object's decomposition to the moduli of its decompositions** the passage from concrete to abstract and back that he regards as constitutive of mathematical creativity?

Things he will catch

- **The thesis title.** *Essai sur les notions de structure et d'existence en mathématiques* — **structure**, not essence. Essence/existence is a different, scholastic pair. First thing he catches.
- **The hinge quote's pagination.** Given as pp. 130–131 in Duffy's Continuum 2011 translation; verify, and be ready for him to cite the Spanish.
- **Which introduction.** The English Continuum volume's introduction is Duffy's. If the project means Zalamea's introduction, name the edition — his Spanish Lautman volume.
- **The hierarchy is the project's.** Not Lautman's, and he will know.
- **"Domination" needs its argument.** Never let it stand without the recurrence form behind it.
- **R4 is not a mixte**, and the project says so in its own records.
- **The Yoneda move is an analogy**, unresourced, and the project cannot make it rigorous any time soon. Say it before he asks; what it would need is a category of worlds, its morphisms, a probe family, and a density or representability theorem.
- **Three claims he may expect that the project does not make**: that sheaf descent proves there is no privileged point of view (it needs equivariance, basepoint-independence, refinement stability and conservative reindexing); that the existence of a contextual transpose supports the adjunction (every adjunction supplies one); that unanimity is sheaf amalgamation.

Reporting movement against his advice

- He urged the big book and ten years of solitary work. Report against that concretely rather than in the abstract.
- What has become stable since: the concurrency side's banked results and the one named postulate; the retirement of cohesive and modal HoTT as the *prescribed* setting for M, with M now to be discovered; the comparison setting as the primary target rather than an absolute claim about one architecture.
- What the Lautman reading changed, in one line: it supplied the justification the construction lacked, and relocated the project's own problem from inside the architecture to the adjunction.